groundplane

HDMI to CSI (RK628H) 30P Board

RK628H 30P Board RevA
11/12/2026 10:55 AM

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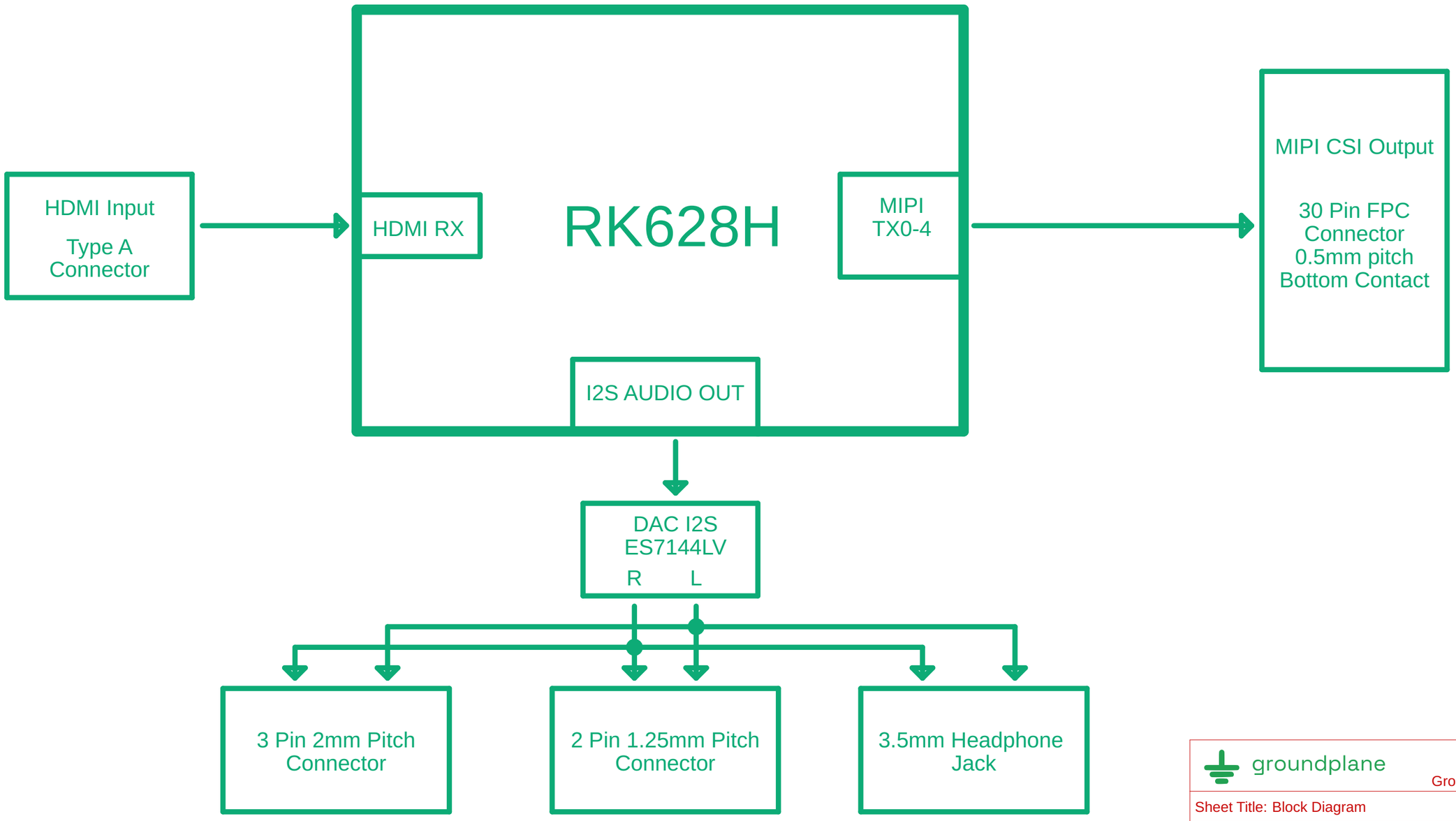
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BLOCK DIAGRAM

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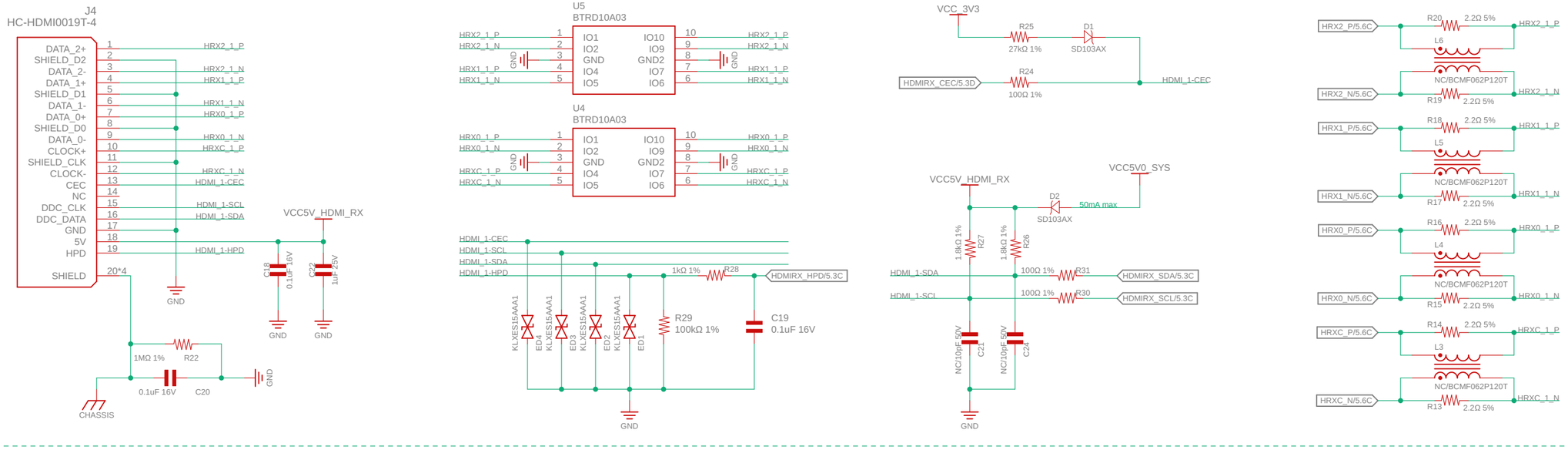
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HDMI INPUT



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Sheet Title: HDMI Input Connector/TVS/Filters	
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HDMI AUDIO

RK628_XTALI — Y5 24MHz 20pF (XIN, XOUT, GND1, GND2) — C13 33pF 50V — C15 33pF 50V — R6 22Ω 1% — RK628_XTALO

U3
RK628H
FOR RK628H NO GVI

VCC1V1_RK628	D8*2	DVDD	
VCC_3V3	H5	VCCIO1	
	E8	VCCIO2_1	
	F8	VCCIO2_2	
VCCA3V3_RK628	M10	PLL_AVDD	
VCCA1V1_RK628	J7	PLL_AVDD_1V1	
	J8	GND30	
VCCA3V3_RK628	C3	GVI/LVDS/MIPI_PLL_AVDD_3V3	
	D5*2	GVI/LVDS/MIPI_AVDD_3V3	
VCCA1V1_RK628	C5*4	GVI/LVDS/MIPI_AVDD_1V1	
	J4	HDMITX_DVDD_1V1	
	J5	HDMITX_DVDD_1V1_2	
	J6	HDMITX_DVDD_1V1_3	
VCCA1V1_RK628	K4	HDMITX_AVDD_3V3	
VCCA3V3_RK628	H9*2	HDMIRX_DVDD_1V1	
	J10	HDMIRX_AVDD_3V3	
	J2	GPIO0_A2/I2S_SCK_M0	
	J1	GPIO0_A3/I2S_LRCK_M0	
	K2	GPIO0_A4/I2S_D0_M0/UARTTX	
	H3	GPIO0_A5/I2S_D1_M0/UARTRX	
	H2	GPIO0_A6/I2S_D2_M0/UARTCTSN	
	H1	GPIO0_A7/I2S_D3_M0/UARTTRTSN	
	L2	GPIO0_B0/HDMITX_HPD	
	K1	GPIO0_B1/HDMITX_SDA	
	L1	GPIO0_B2/HDMITX_SCL	
	M2	GPIO0_B3/HDMITX_CEC	
	H7	GPIO1_A0/TEST_CLKO	
	H8	GPIO1_A1/SFC_CSLK	
	K8	GPIO1_B0/HDMIRX_HPD_M0	
	M8	GPIO1_B1/HDMIRX_SDA_M0	
	L8	GPIO1_B2/HDMIRX_SCL_M0	
	K9	GPIO1_B3/HDMIRX_CEC_M0	
	G10	GPIO2_A0/VOP_D0	
	G11	GPIO2_A1/VOP_D1	
	G12	GPIO2_A2/VOP_D2	
	F9	GPIO2_A3/VOP_D3	
	F12	GPIO2_A4/VOP_D4	
	F11	GPIO2_A5/VOP_D5	
	F10	GPIO2_A6/VOP_D6	
	E10	GPIO2_A7/VOP_D7	
	E12	GPIO2_B0/VOP_D8	
	E11	GPIO2_B1/VOP_D9	
	D10	GPIO2_B2/VOP_D10	
	D11	GPIO2_B3/VOP_D11	
	C11	GPIO2_B4/VOP_D12	
	D12	GPIO2_B5/VOP_D13	
	C12	GPIO2_B6/VOP_D14	
	B12	GPIO2_B7/VOP_D15	
	B11	GPIO2_C0/VOP_D16/XIPSFCCSN	
	A11	GPIO2_C1/VOP_D17/XIPSFCMISO	
	C10	GPIO2_C2/VOP_D18/XIPSFCMOSI	
	B10	GPIO2_C3/VOP_D19/JTAGTDO	
	A10	GPIO2_C4/VOP_D20/JTAGTDI	
	D9	GPIO2_C5/VOP_D21/JTAGTMS	
	C9	GPIO2_C6/VOP_D22/JTAGTCK	
	C8	GPIO2_C7/VOP_D23/JTAGTRSN	
	A1*29	VSS	

	M9	OSC_IN	RK628_XTALI
	L9	OSC_OUT	RK628_XTALO
	H4	RESETN	RK628_RSTN_3V3/7.1C
	D7	TEST	GND
	C7	I2C_ADDR	
	L3	I2C_SCL	I2C_SCL_3V3/7.1D
	K3	I2C_SDA	I2C_SDA_3V3/7.1E
	J3	INT/SPIBOOT	RK628_INTN_3V3/7.1F
	A9	VOP_DCLK	
	B9	GPIO3_A0/VOP_DEN	
	A8	GPIO3_A1/VOP_HSYNC	
	B8	GPIO3_A3/VOP_VSYNC	
	A7	H-NC1/GPIO3_B1/GVI_HPD	
	B7	H-NC2/GPIO3_B2/GVI_LOCK	
	G9	HDMIRX_EXTR	R34 2kΩ 1% GND
	L11	HDMIRX_CLKN	HRXC_N/4.8D
	L12	HDMIRX_CLKP	HRXC_P/4.8D
	K11	HDMIRX_D0N	HRX0_N/4.8C
	K12	HDMIRX_D0P	HRX0_P/4.8C
	J11	HDMIRX_D1N	HRX1_N/4.8C
	J12	HDMIRX_D1P	HRX1_P/4.8C
	H11	HDMIRX_D2N	HRX2_N/4.8B
	H12	HDMIRX_D2P	HRX2_P/4.8B
	L4	HDMITX_CLKN	
	M4	HDMITX_CLKP	
	L5	HDMITX_D0N	
	M5	HDMITX_D0P	
	L6	HDMITX_D1N	
	M6	HDMITX_D1P	
	L7	HDMITX_D2N	
	M7	HDMITX_D2P	
	M3	HDMITX_EXTR	
	B1	GVI/LVDS/MIPI_REXT	R11 4.02kΩ 1% GND
	G1	GVI/LVDS/MIPI_TX0N	MIPI-CSI0-D0_N/7.8D
	G2	GVI/LVDS/MIPI_TX0P	MIPI-CSI0-D0_P/7.8D
	F1	GVI/LVDS/MIPI_TX1N	MIPI-CSI0-D1_N/7.8D
	F2	GVI/LVDS/MIPI_TX1P	MIPI-CSI0-D1_P/7.8D
	E1	GVI/LVDS/MIPI_TX2N	MIPI-CSI0-CK_N/7.8D
	E2	GVI/LVDS/MIPI_TX2P	MIPI-CSI0-CK_P/7.8D
	D1	GVI/LVDS/MIPI_TX3N	MIPI-CSI0-D2_N/7.8D
	D2	GVI/LVDS/MIPI_TX3P	MIPI-CSI0-D2_P/7.8D
	C1	GVI/LVDS/MIPI_TX4N	MIPI-CSI0-D3_N/7.8D
	C2	GVI/LVDS/MIPI_TX4P	MIPI-CSI0-D3_P/7.8D
	B2	GVI/LVDS/MIPI_TX5N	
	A2	GVI/LVDS/MIPI_TX5P	
	B3	GVI/LVDS/MIPI_TX6N	
	A3	GVI/LVDS/MIPI_TX6P	
	B4	GVI/LVDS/MIPI_TX7N	
	A4	GVI/LVDS/MIPI_TX7P	
	B5	GVI/LVDS/MIPI_TX8N	
	A5	GVI/LVDS/MIPI_TX8P	
	B6	GVI/LVDS/MIPI_TX9N	
	A6	GVI/LVDS/MIPI_TX9P	

TP1 TP2 TP3 UART Access for debug — R35 100Ω 1% — C51 0.1uF 16V

TP4 — C5 0.1uF 16V — R12 10kΩ 1%

I2S_HDMI_SCLK/6.6B — R40 33Ω 1%
I2S_HDMI_LRCK/6.6B — R41 33Ω 1%
I2S_HDMI_SDI/6.6B — R42 33Ω 1%
I2S_HDMI_MCLK/6.6C — R43 33Ω 1%

HDMIRX_HPD/4.5C — R32 22Ω 1%
HDMIRX_SDA/4.6C — R37 22Ω 1%
HDMIRX_SCL/4.6C — R36 22Ω 1%
HDMIRX_CEC/4.6B — R33 22Ω 1%

C3/D5/E4
VCCA3V3_RK628 — C26 10uF 10V — C29 0.1uF 16V — CH1 0.1uF 16V

J10
VCCA3V3_RK628 — C23 10uF 10V — C25 0.1uF 16V

D4/E3/C5/F3
VCCA1V1_RK628 — C33 10uF 10V — C34 0.1uF 16V — C31 0.1uF 16V — C32 0.1uF 16V

H9/J9
VCCA1V1_RK628 — C28 10uF 10V — C27 0.1uF 16V

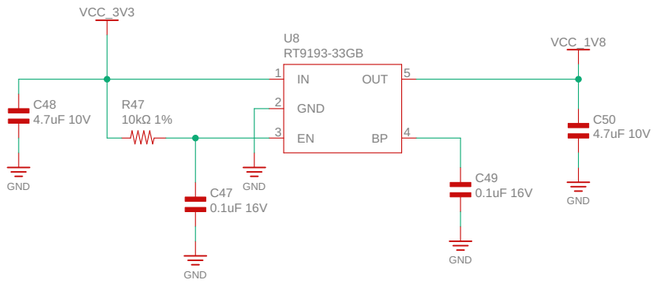
D8/E9
VCC1V1_RK628 — C30 0.1uF 16V

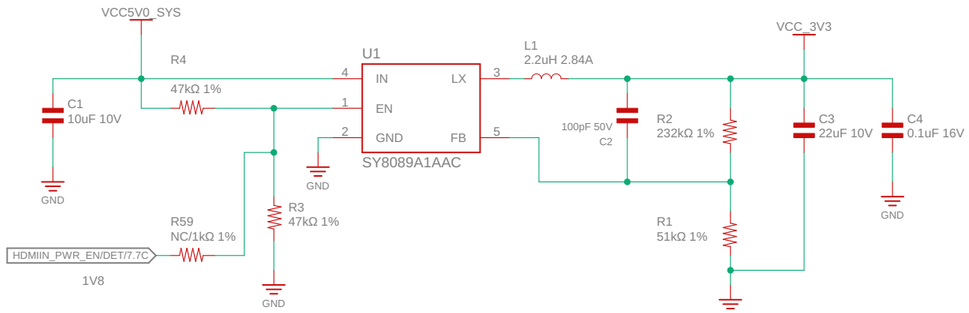
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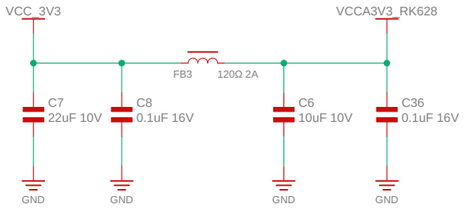
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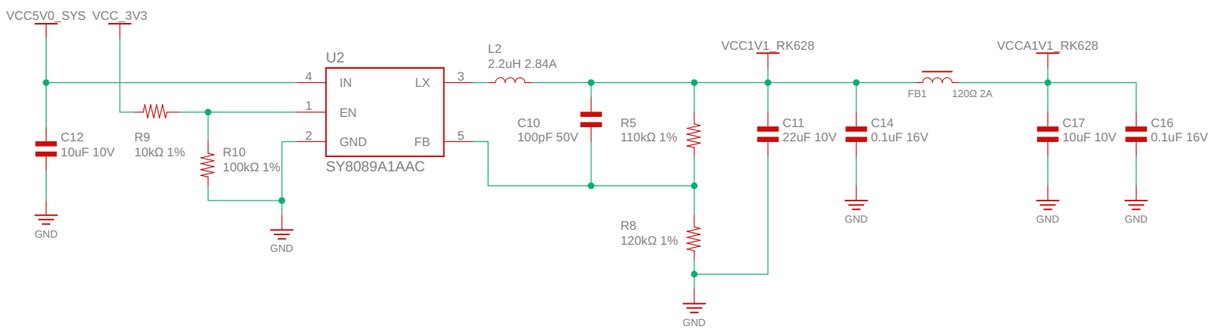
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POWER

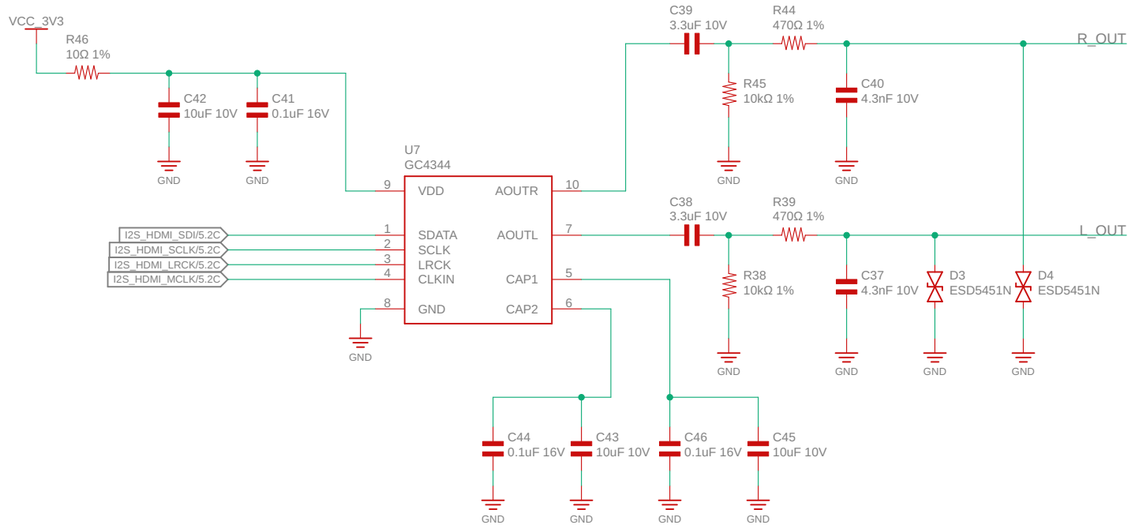


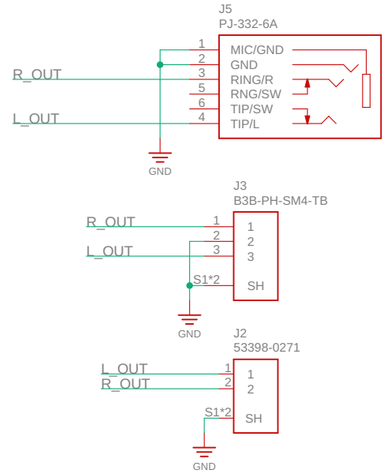






AUDIO





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MNT1
MOUNTHOLE_M2.5_4.8MM_PAD
MNT2
MOUNTHOLE_M2.5_4.8MM_PAD
MNT3
MOUNTHOLE_M2.5_4.8MM_PAD
MNT4
MOUNTHOLE_M2.5_4.8MM_PAD
CHASSIS
1MΩ 1%
R66
0.1uF 16V
C53
GND

VCC_1V8
R23
4.7kΩ 1%
VCC5V_HDMI_RX
HDMIRX_DETN
R21
10kΩ 1%
Q7
S8050
R58
51kΩ 1%
GND

VCC_3V3
R57
10kΩ 1%
R56
10kΩ 1%
RK628_RSTN_3V3/5.7B
Q6
2SK3018
HDMIIN_RSTN
2SK3018
Q5
GND

HDMIIN_PWR_EN/DET
R64
10kΩ 1%
R65
30kΩ 1%
U9
NC/ME2808A33XG
VIN
VOUT
GND
R62
0Ω 1%
R63
NC/100kΩ 1%
GND

HDMIIN_INTN
R60
NC/0Ω 1%
SW1/MIPI_PWR0_CAM_OR_5V

J1
FPC30-T1T1-2021-A
SDA_CAM_1V8
SCL_CAM_1V8
HDMIIN_PWR_EN/DET/6.1D
HDMIIN_RSTN
SW1/MIPI_PWR0_CAM_OR_5V
HDMIIN_INTN
HDMIRX_DETN
SW2
MIPI-CSI0-D0_P/5.6D
MIPI-CSI0-D0_N/5.6D
MIPI-CSI0-D1_P/5.6D
MIPI-CSI0-D1_N/5.6D
MIPI-CSI0-CK_P/5.6D
MIPI-CSI0-CK_N/5.6D
MIPI-CSI0-D2_P/5.6D
MIPI-CSI0-D2_N/5.6D
MIPI-CSI0-D3_P/5.6D
MIPI-CSI0-D3_N/5.6D
VCC5V0_SYS
31*2
SH
GND

VCC_3V3
VCC_1V8
R48
2.2kΩ 1%
Q2
2SK3018
R7
NC/2.2kΩ 1%
I2C_SCL_3V3/5.6B
SCL_CAM_1V8
R49
NC/0Ω 1%

VCC_3V3
VCC_1V8
R51
2.2kΩ 1%
2SK3018
Q3
R50
NC/2.2kΩ 1%
I2C_SDA_3V3/5.6B
SDA_CAM_1V8
R52
NC/0Ω 1%

VCC_3V3
VCC_1V8
R54
2.2kΩ 1%
Q4
2SK3018
R53
NC/2.2kΩ 1%
RK628_INTN_3V3/5.6B
HDMIIN_INTN
R55
NC/0Ω 1%

VCC5V0_SYS
U6
SGM3157YC6
IN
NO
V+
GND
COM
NC
SW2
SW1/MIPI_PWR0_CAM_OR_5V
HDMIIN_INTN
R61
NC/1MΩ 1%
C9
0.1uF 16V
GND

DET	SW1	SW2	INTH
5V	5V	IO	SW2
IO	IO	GND	SW1

VCC5V0_SYS
C52
10uF 10V
C35
0.1uF 16V
GND

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